



Yakeen Plus (2025)

Practice Test- 09

DURATION :200 Minutes

DATE : 20/11/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (1)
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CHEMISTRY

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ZOOLOGY

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200. (3)

SECTION-(I) PHYSICS

1. (1)

Iron (Fe) is the most stable nucleus.

From the concept of binding energy, the binding energy per nucleon increases on an average and reaches a maximum of about 8.7 MeV per nucleon for $A = 50 - 80$ and generally a peak is found for ${}_{26}Fe^{56}$, which is an iron isotope. This is also the most stable nucleus from all of them because it has even number of protons (26) and even number of neutrons (30), i.e., even-even pair.

[New NCERT Class 12th Page No. 390]

2. (1)

Concept of threshold frequency.

[New NCERT Class 12th Page No. 389]

3. (3)

$$h\nu = h\nu_0 + KE_{\max}$$

$$KE_{\max} = h\nu - h\nu_0$$

$$V_0 = \frac{h}{e}(\nu - \nu_0)$$

[New NCERT Class 12th Page No. 395]

4. (1)

Photocurrent $\propto$ Intensity of light

[New NCERT Class 12th Page No. 375]

5. (3)

$$\phi = \phi_0 + KE$$

$$KE_i = \frac{1}{2}mV^2$$

$$KE_f = 9\left(\frac{1}{2}mV^2\right) = 9KE_i$$

$$\phi_i = h\nu = \phi$$

$$\phi_f = 2h\nu = 2\phi$$

$$\phi = \phi_0 + KE$$

$$2\phi = \phi_0 + 9KE$$

$$2\phi = \phi_0 + 9\phi - 9\phi_0$$

$$7\phi = 8\phi_0$$

$$\phi_0 = \frac{7h\nu}{8}$$

[New NCERT Class 12th Page No. 391]

6. (4)

$$\text{Energy of incident light} = \frac{12375}{\lambda} \text{ \AA}$$

$$= \frac{12375}{3000} = 4.125 \text{ eV}$$

$$K_{\max} = E - w_0 = 4.125 - 1 = 3.125 \text{ eV}$$

$$\frac{1}{2}mv^2 = 3.125 \times 1.6 \times 10^{-19}$$

$$\Rightarrow v = \sqrt{\frac{2 \times 3.125 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 10^6$$

[New NCERT Class 12th Page No. 396]

7. (1)

$$\text{Energy of photon, } E = h\nu = \frac{hc}{\lambda_p}$$

$$\lambda_p = \frac{hc}{E}$$

$$\text{Wavelength of electron, } \lambda_e = \frac{h}{\sqrt{2mE}}$$

$$\therefore \frac{\lambda_p}{\lambda_e} = \frac{hc}{E} \times \frac{\sqrt{2mE}}{h} = c\sqrt{\frac{2m}{E}}$$

[New NCERT Class 12th Page No. 392]

8. (1)

Same stopping potential means incident light is of same frequency.

[New NCERT Class 12th Page No. 392]

9. (1)

In fission process energy is released by converting some mass into energy. Hence, mass of fission product < mass of parent nucleus.

[New NCERT Class 12th Page No. 453]

10. (1)

Fact.

[New NCERT Class 12th Page No. 455]

11. (3)

Mass defect = Δm

$$\Delta m = (A - Z)m_n + Zm_p$$

A = Atomic mass number

Z = Atomic number

[New NCERT Class 12th Page No. 443]

12. (2)

$$BE = \Delta mc^2$$

Δm = mass defect in kg

$$\Delta m = (17 - 8)M_n + 8M_p - M_0$$

$$BE = (9M_n + 8M_p - M_0)C^2$$

[New NCERT Class 12th Page No. 443]

13. (3)

In heavy nuclei $BE/\text{nucleon}$ is low so it is less stable and lighter nuclei $BE/\text{nucleon}$ is high so it is highly stable. Hence option (3) is correct.

[New NCERT Class 12th Page No. 456]

14. (3)

Sum of atomic number and mass number at reactant and product remain constant

$$13 + 0 = 11 + Z$$

$$Z = 2$$

$$27 + 1 = A + 24$$

$$A = 4$$

Particle is alpha.

[New NCERT Class 12th Page No. 447]

15. (2)

$$\text{Total B.E. of product} = 4x_2$$

$$\text{Total B.E. of reactant} = 4x_1$$

$$\text{Energy released } Q = 4(x_2 - x_1)$$

$$Q = 4(x_2 - x_1)$$

[New NCERT Class 12th Page No. 447]

16. (1)

It is so because when mass number increases then number of protons also increases due to which repulsion increase which further decrease the binding energy per nucleon. Also, number of nucleon increases which decreases B.E. per nucleon value.

[New NCERT Class 12th Page No. 444]

17. (1)

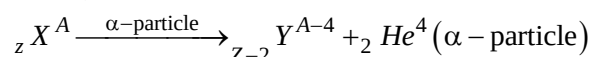
Isotones have same neutron numbers but different atomic number.

[New NCERT Class 12th Page No. 448]

18. (3)

The emission of an α -particle from the atom of an element reduces its atomic number by 2, and mass number by 4.

Hence, the radioactive emission is as follows:



Also from law of conservation of momentum,

$$m \times 0 = m_y v_y + m_\alpha v_\alpha$$

[$\because$ nucleus is originally at rest, $v = 0$]

$$= (A - 4) v_y + 4v$$

$$\Rightarrow v_y = -\frac{4v}{A-4}$$

[New NCERT Class 12th Page No. 446]

19. (4)

$$Q = 2(4 \times 7.06) - (7 \times 5.60)$$

$$Q = 17.3 \text{ MeV}$$

[New NCERT Class 12th Page No. 443]

20. (3)

This is because hydrogen nucleus contains only one nucleon.

[New NCERT Class 12th Page No. 444]

21. (4)

$$\frac{\lambda_{\alpha}}{\lambda_{\text{deuteron}}} = \frac{m_{\text{deuteron}}}{m_{\alpha}} = 1:2$$

[New NCERT Class 12th Page No. 398]

22. (1)

$$\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{m}}$$

[New NCERT Class 12th Page No. 398]

23. (3)



[New NCERT Class 12th Page No. 429]

24. (2)

For 7th excited state $n = 8$

$$\text{Number of lines} = \frac{n(n-1)}{2} = \frac{8(8-1)}{2} = 4 \times 7 = 28$$

[New NCERT Class 12th Page No. 426]

25. (2)

$$V \propto \frac{1}{n}$$

$$\text{And } r \propto n^2$$

$$\frac{r}{V} \propto n^3$$

[New NCERT Class 12th Page No. 422]

26. (3)

$$B = \frac{\mu_0 I}{2r}$$

$$B \propto \frac{I}{r}$$

$$B \propto \frac{e}{Tr}$$

$$\therefore B \propto \frac{Z^3}{n^5}$$

[New NCERT Class 12th Page No. 423]

27. (3)

$$\lambda = \frac{1}{R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}$$

$$\lambda' = \frac{1}{R\left(\frac{1}{3^2} - \frac{1}{4^2}\right)}$$

$$\frac{\lambda'}{\lambda} = \frac{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\left(\frac{1}{3^2} - \frac{1}{4^2}\right)} \Rightarrow \lambda' = \frac{20\lambda}{7}$$

[New NCERT Class 12th Page No. 428]

28. (3)

As per conservation of momentum

$$\therefore \frac{h}{\lambda} = mv$$

$$\therefore v = \frac{h}{\lambda m} = \frac{h}{m} RZ^2 \left[\frac{1}{1^2} - \frac{1}{5^2} \right]$$

$$\therefore v = \frac{hR}{25} \frac{24}{m}$$

[New NCERT Class 12th Page No. 429]

29. (4)

$$K = |E|$$

[New NCERT Class 12th Page No. 428]

30. (3)

$$L = \frac{nh}{2\pi}$$

$$L \propto n \propto \sqrt{r}$$

[New NCERT Class 12th Page No. 422]

31. (2)

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$f = Rc\left(\frac{1}{4} - \frac{1}{9}\right)$$

$$f = \frac{Rc5}{36} = \frac{5Rc}{36}$$

[New NCERT Class 12th Page No. 428]

32. (4)

$$L = \frac{nh}{2\pi}$$

[New NCERT Class 12th Page No. 422]

33. (2)

$$\frac{1}{\lambda_L} = R\left[\frac{1}{1^2} - \frac{1}{\infty}\right] = R$$

$$\frac{1}{\lambda_B} = R\left[\frac{1}{2^2} - \frac{1}{\infty}\right] = \frac{R}{4}$$

$$\frac{1}{\lambda_P} = R\left[\frac{1}{3^2} - \frac{1}{\infty}\right] = \frac{R}{9}$$

$$\lambda_L : \lambda_B : \lambda_P = \frac{1}{R} : \frac{4}{R} : \frac{9}{R} = 1 : 4 : 9$$

[New NCERT Class 12th Page No. 428]

34. (4)

$$\frac{1}{\lambda_B} = R\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$= R\left[\frac{1}{4} - \frac{1}{9}\right] = \frac{5R}{36}$$

Wavelength of second line of Lyman series.

$$\frac{1}{\lambda_L} = R\left[\frac{1}{1^2} - \frac{1}{9}\right] = \frac{8R}{9}$$

$$\frac{\lambda_B}{\lambda_L} = \frac{32}{5}$$

[New NCERT Class 12th Page No. 422]

35. (3)

$$\frac{h}{2\pi}$$

[New NCERT Class 12th Page No. 422]

36. (3)

The neutron has no net charge and mass is slightly larger than proton. Protons and neutrons constitute the nuclei of atoms. Outside the nucleus, the neutron is unstable. The life of neutron outside the nucleus is about 932 seconds.

[New NCERT Class 12th Page No. 454]

37. (1)

$$|-4V| > |-2V|$$

[New NCERT Class 12th Page No. 391]

38. (2)

Photoelectric effect demonstrates the particle nature of light. Number of emitted photoelectrons depends upon the intensity of light.

[New NCERT Class 12th Page No. 392]

39. (3)

$$\lambda = \frac{h}{\sqrt{2mQV}} \Rightarrow \lambda \propto \frac{1}{\sqrt{mQ}} \Rightarrow \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha Q_\alpha}{m_p Q_p}}$$

$$= \sqrt{\frac{4m_p \times 2Q_p}{m_p \times Q_p}} = 2\sqrt{2}$$

[New NCERT Class 12th Page No. 398]

40. (1)

$$\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$$

[New NCERT Class 12th Page No. 398]

41. (1)

$$n = 2 \text{ ————— } E_2 = -\frac{13.6}{(2)^2} = -3.4 \text{ eV}$$

$$n = 1 \text{ ————— } \uparrow \text{ ————— } E_1 = -13.6 \text{ eV}$$

$$E_{1 \rightarrow 2} = -3.4 - (-13.6) = +10.2 \text{ eV}$$

[New NCERT Class 12th Page No. 422]

42. (4)

- Most of the atom is hollow.
- The whole mass of the atom is concentrated in a small centre called nucleus.
- Nucleus is positively charged.

[New NCERT Class 12th Page No. 415]

43. (4)

$$2E - E = \frac{hc}{\lambda} \Rightarrow E = \frac{hc}{\lambda}$$

$$\frac{4E}{3} - E = \frac{hc}{\lambda'} \Rightarrow \frac{E}{3} = \frac{hc}{\lambda'} \therefore \frac{\lambda'}{\lambda} = 3 \Rightarrow \lambda' = 3\lambda$$

[New NCERT Class 12th Page No. 416]

44. (3)

$$\text{Emitted energy } \Delta E = \frac{hc}{\lambda} \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

[New NCERT Class 12th Page No. 422]

45. (4)

In 1913, Niels Bohr proposed a model for the hydrogen atom. The Bohr model was based on the following assumptions.

- The electron in a hydrogen atom travels around the nucleus in a circular orbit.
- The nucleus is of infinite mass and is rest.
- Electrons in a quantized orbit will not radiate energy.
- Mass of electron remains constant.

[New NCERT Class 12th Page No. 422]

46. (3)

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\lambda \propto \frac{1}{Z^2}, \lambda_{\text{Li}^{++}} : \lambda_{\text{He}^+} : \lambda_{\text{H}} = 4 : 9 : 36$$

[New NCERT Class 12th Page No. 422]

47. (2)

$$r \propto n^2 \Rightarrow \frac{r_{(n=2)}}{r_{(n=3)}} = \frac{4}{9} \Rightarrow r_{(n=3)} = \frac{9}{4}R = 2.25R$$

[New NCERT Class 12th Page No. 428]

48. (4)

Lyman series lies in the UV region.

[New NCERT Class 12th Page No. 428]

49. (1)

$$\text{P.E.} \propto -\frac{1}{r} \text{ and K.E.} \propto \frac{1}{r}$$

As r increases so K.E. decreases but P.E. increases.

[New NCERT Class 12th Page No. 428]

50. (1)

When two lighter nuclei moving at very high speeds fuse together to form a single heavier nucleus then this phenomenon is called nuclear fusion. The mass of the nucleus obtained after fusion is less than the sum of the masses of the nuclei which are fused together. The lost mass is obtained in the form of energy.

[New NCERT Class 12th Page No. 415]

SECTION-(II) CHEMISTRY

51. (3)
Group-13 elements show +1 and +3 oxidation states. Tendency to show +3 oxidation state decreases down the group.
52. (1)
Nitrogen molecule is diatomic containing a triple bond between two N atoms, $\text{N} \equiv \text{N}$. Therefore, nitrogen molecule is formed by sharing six electrons.
53. (1)
Because of their smaller size and absence of d-orbitals, the first members of p-block elements possess different properties from the remaining members of their corresponding group.
54. (4)
AgF is highly soluble in H_2O therefore, NaF formed in the Lassaigne's extract does not produce a precipitate with AgNO_3 .
55. (1)
Amongst the elements of group 13, B has the highest sum of first three ionisation enthalpies. As a result, it has a little tendency to lose electrons and hence is least electropositive amongst group 13 elements.
56. (3)
Down the group, +1 oxidation state becomes more and more stable as compared to +3 oxidation state due to inert pair effect.
57. (1)
The decrease in ionization enthalpy from C to Sn is due to increase in atomic size and screening effect of the inner electrons which outweigh the effect of increased nuclear charge. However, the small increase in ionization enthalpy from Sn to Pb is due to a considerable increase in nuclear charge which outweighs the shielding effect of all the electrons in the inner shells including those of 4f and 5d - electrons.
58. (2)
Metallic character increases as we go down the group. Carbon and silicon are non-metals, germanium is metalloid and tin and lead are distinctly metals.
59. (2)
The solution contains ions of zinc.
60. (3)
Nitrogen form N_2 (i.e. $\text{N} \equiv \text{N}$) but phosphorus form P_4 , because in P_2 , $p_\pi - p_\pi$ bonding is present which is a weaker bonding.
61. (4)
The electronic configuration of 'Al' is:
 $_{13}\text{Al} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^1$.
The most stable oxidation state of aluminium is +3.
62. (4)
 CdS and As_2S_3
63. (3)
Xenon is the only noble gas that readily forms compounds with fluorine. It forms a variety of compounds such as xenon hexafluoride (XeF_6) and xenon tetrafluoride (XeF_4). Noble gases like helium, neon, and argon have very stable electron configurations and typically do not form compounds under normal conditions.
64. (4)
A lead chloride original solution (i.e., O.S) on mixing with K_2CrO_4 solution gives a yellow precipitate soluble in aqueous sodium hydroxide.
 $\text{Pb}^{2+} + \text{CrO}_4^{2-} \rightarrow \text{PbCrO}_4 \downarrow (\text{yellow}),$
 $\text{PbCrO}_4 \downarrow + 4\text{OH}^- \rightarrow \text{Pb}(\text{OH})_4^{2-} + \text{CrO}_4^{2-}$
65. (1)
Presence of 10 d-electrons in gallium which have poor screening effect on the outer electrons is the reason for smaller size of gallium. This increases the force of attraction between the outermost electrons and nucleus resulting in decrease in atomic radius.
66. (2)
Concentrated acid group: NO_3^- , CH_3COO^- , Cl^- , Br^- , I^- .

67. (4)
The acidic solution of a salt produced a deep blue colour with starch iodide solution. The salt may be nitrite. Acidic solution of nitrite first forms nitrous acid which decomposes to NO, which is then oxidised to nitrogen dioxide.

$$2\text{NaNO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HNO}_2$$

$$3\text{HNO}_2 \rightarrow \text{HNO}_3 + 2\text{NO} \uparrow + \text{H}_2\text{O}$$

$$2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$$
Nitrogen dioxide gas decomposes iodide to free iodine which gives blue colour with starch solution.
68. (4)
On moving down the group 13 from B to Al, $\Delta_i H_1$ decreases due to increase in atomic size and screening effect. $\Delta_i H_1$ of Ga is slightly higher than that of Al while that of Tl is much higher than those of Al, Ga and In, due to poor shielding effect of d-electrons in case of Ga and poor shielding effect of d- and f-electrons in case of Tl.
69. (2)
C is strictly non-metallic. Si is essentially a, non-metal. Ge is a semimetal (metalloid) whereas Sn and Pb are soft metals with low melting points.
70. (1)
Order of dipole moment decreases as
 $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$
(Based upon electronegativity)
71. (2)
Noble gases have high ionization energies because of their stable electron configuration. They are chemically inert due to their full valence electron shells, which makes it difficult to remove an electron. Their boiling points are low, and they typically do not readily form covalent bonds with halogens.
72. (1)
 AgNO_3 is often called Lunar caustic since it creates a burning sensation such as caustic and leaves stain similar to moon luna (black stain while decomposition to metallic silver). This is the reason it is stored in coloured bottles. Nitrate of silver on strong heating leaves the metal as the residue.
73. (2)
 $\text{Si}^{4+} > \text{Ge}^{4+} > \text{Sn}^{4+} > \text{Pb}^{4+}$. Stability of M^{4+} decreases down the group due to increase in inert pair effect down the group.
74. (3)
Ca, Ba, and Sr ions are precipitated in fifth group as their carbonates. Group reagent for V group is ammonium carbonate in the presence of NH_4Cl and NH_4OH .
75. (4)
Because of its small size, nitrogen has a strong tendency to form $p\pi-p\pi$ bonds. It has high electronegativity and does not have any d-orbital. Also, it has high ionisation enthalpy due to its stable half-filled electronic configuration.
76. (1)
As the electronegativity decreases as we move down the group, the tendency of group 16 elements to show -2 oxidation state decreases from S to Po.
77. (3)
Neon is commonly used in neon signs because it emits a bright red-orange glow when an electric current is passed through it. Other noble gases like argon and krypton are used in different types of lighting, but neon is particularly famous for neon signs.
78. (4)
 Co^{2+} ion will not be precipitated by H_2S in the presence of dil. HCl as dil. HCl will suppress the ionization of H_2S .
To affect the precipitation, H_2S should be used in the ammoniacal solution.
79. (1)
Carbon forms a number of catenated halides.
80. (2)
Brown ring test is done for the confirmation of NO_3^- ions.
81. (3)

$$\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$$

$$\text{ZnCl}_2 + 2\text{NaOH} \rightarrow \text{Zn}(\text{OH})_2 + 2\text{NaCl}$$
white ppt.

$$\text{Zn}(\text{OH})_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + 2\text{H}_2\text{O}.$$

82. (2)
When silver nitrate (AgNO_3) solution is added to a solution containing chloride ions (Cl^-), a white precipitate of silver chloride (AgCl) forms:
 $\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl (s)}$
83. (1)
Sulphate ions (SO_4^{2-}) can be confirmed by adding barium chloride (BaCl_2) solution. A white precipitate of barium sulphate (BaSO_4) forms if sulphate ions are present:
 $\text{SO}_4^{2-} + \text{Ba}^{2+} \rightarrow \text{BaSO}_4 (\text{s})$
84. (3)
Pb due to inert pair effect shows most stable +2 oxidation state.
85. (2)
Lower oxidation state become more stable on moving down the group
 $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
86. (2)
Stability of +3 oxidation state decreases from Al to Tl. B always shows +3 oxidation state in all of its compounds.
87. (2)
NO reacts with FeSO_4 to form a brown ring complex, $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$.
88. (3)
Aluminium ions (Al^{3+}) can be confirmed by adding ammonium hydroxide (NH_4OH) and ammonium chloride (NH_4Cl) buffer solution. A white gelatinous precipitate of aluminium hydroxide ($\text{Al}(\text{OH})_3$) forms, which is soluble in excess ammonium hydroxide, forming a colorless solution.
89. (2)
The increasing order of atomic radii of group 13 elements is $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$.
Atomic radius of Ga is slightly lower than that of Al due to the presence of d-electrons in Ga which do not shield the nucleus effectively.
90. (1)
The mixture contains NH_4^+ ions.
91. (1)
The brick's red colour is produced by calcium metal in flame. It is caused by the thermal energy's excitation of electrons, which is followed by the de-excitation of an electron to its ground state and the release of visible light.
92. (4)
Due to small size of fluorine atoms, electrons repel each other and hence, bond dissociation enthalpy of F_2 is low.
93. (4)
Because of high bond energy of $\text{N} \equiv \text{N}$ bond, N_2 is chemically inert.
94. (2)
The third group solution should be treated with excess of NH_4Cl .
95. (3)
Electronic configuration of Al - $1s^2 2s^2 2p^6 3s^2 3p^1$
Electronic configuration of Mg - $1s^2 2s^2 2p^6 3s^2$
It is difficult to remove electron from paired $3s^2$ -orbital than unpaired $3p^1$. Hence, *I.E.* of Al is lower than Mg.
96. (2)
The water will be coloured by vapours evolved from the test tube containing bromide.
 $\text{Br}_2 + \text{H}_2\text{O} \rightarrow \text{Brown Colour}$
 $\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{Colourless}$
 Br_2 impart a brown color to the aqueous solution while NO_2 does not.
97. (3)
Fluorine and chlorine having low van der Waals forces of attraction are gases. Br_2 is liquid, while iodine with high molecular mass and more van der Waals force of attraction between the molecules is solid at room temperature.
98. (3)
Ionization energy decreases from fluorine to iodine due to increase in size.
99. (1)
Electron affinity decreases down the group as atomic size increases.
100. (1)
Nitrogen exhibits a large number of oxidation states from -3 to +5.

SECTION-(III) BOTANY

101. (2)

The current species extinction rates are estimated to be 100 to 1,000 times faster than in the pre-human times and our activities are responsible for the faster rates.

[New NCERT Class 12th Page No. 222]

102. (2)

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee. One petal of its flower bears an uncanny resemblance to the female of the bee in size, colour and markings. The male bee is attracted to what it perceives as a female, 'pseudocopulates' with the flower, and during that process is dusted with pollen from the flower.

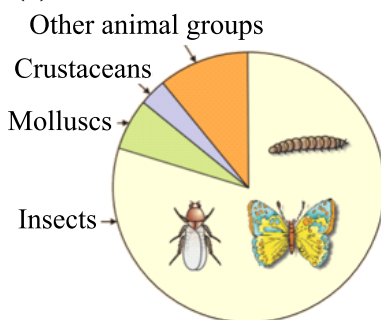
[New NCERT Class 12th Page No. 202]

103. (3)

- The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro.
- The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

[New NCERT Class 12th Page No. 225]

104. (3)



[New NCERT Class 12th Page No. 218]

105. (3)

The Sarguja, Chanda and Bastar areas are in Madhya Pradesh.

[New NCERT Class 12th Page No. 225]

106. (4)

Three of the hotspots – Western Ghats and Sri Lanka, Indo-Burma and Himalaya – cover our country's exceptionally high biodiversity regions.

[New NCERT Class 12th Page No. 224]

107. (4)

Endemism	Species found in a particular area only
Hotspot of India	Western Ghats
Sacred grove	Khasi and Jaintia hills
Cryopreservation	Advanced <i>ex-situ</i> Conservation

[New NCERT Class 12th Page No. 224, 225]

108. (4)

The 'Evil Quartet' of biodiversity loss includes:

- Habitat loss
- Introduction of alien species
- Over-exploitation
- Co-extinction

[New NCERT Class 12th Page No. 222, 223]

109. (2)

The slope of the line is much steeper when the species-area relationships among very large areas are considered.

[New NCERT Class 12th Page No. 220]

110. (3)

Rivets on the wings represent key species.

[New NCERT Class 12th Page No. 221]

111. (4)

- The accelerated rates of species extinctions that the world is facing now are largely due to human activities.
- The most dramatic examples of habitat loss come from tropical rain forests.
- Ecologists warn that if the present trends continue, nearly half of all the species on earth might be wiped out within the next 100 years.
- Amazon rain forest is known for having the highest biodiversity in the world.

[New NCERT Class 12th Page No. 222]

112. (3)

- In India, ecologically unique and biodiversity-rich regions are legally protected as biosphere reserves, national parks and sanctuaries.
- India has a history of religious and cultural traditions that emphasised protection of nature.

[New NCERT Class 12th Page No. 225]

113. (4)

- In a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFC.
- Detritus food chain may be connected with the grazing food chain at some levels: some of the organisms of DFC are prey to the GFC animals, and in a natural ecosystem, some animals like cockroaches, crows, etc., are omnivores.

[New NCERT Class 12th Page No. 210]

114. (1)

A sparrow is a primary consumer when it eats seeds, fruits, peas, and a secondary consumer when it eats insects and worms.

[New NCERT Class 12th Page No. 213]

115. (3)

Each trophic level has a certain mass of living material at a particular time called as the standing crop.

[New NCERT Class 12th Page No. 211]

116. (2)

If in a pond there are 20 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 28, we calculate the birth rate as $8/20 = 0.4$ offspring per lotus per year.

[New NCERT Class 12th Page No. 191]

117. (2)

Emigration is the number of individuals of the population who left the habitat and gone elsewhere during the time period under consideration.

[New NCERT Class 12th Page No. 193]

118. (4)

Cuscuta and Mango tree	Parasitism
Cuckoo and crow	Brood parasitism
Sea anemone and clown fish	Commensalism
Female wasp and fig	Mutualism

[New NCERT Class 12th Page No. 200, 201]

119. (2)

The vertical distribution of different species occupying different levels is called stratification.

[New NCERT Class 12th Page No. 206]

120. (4)

- It represents pyramid of energy.
- It is always upright.
- It follows 10% energy law transfer.

[New NCERT Class 12th Page No. 213]

121. (2)

The primary consumers are herbivores.

[New NCERT Class 12th Page No. 209]

122. (2)

Forest, grassland and desert are some examples of terrestrial ecosystems; pond, lake, wetland, river and estuary are some examples of aquatic ecosystems.

[New NCERT Class 12th Page No. 205]

123. (4)

- Gross primary productivity is always greater than the net primary productivity.
- Primary productivity depends on the plant species inhabiting a particular area.
- It also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
- A constant input of solar energy is the basic requirement for any ecosystem to function and sustain.

[New NCERT Class 12th Page No. 206, 207]

124. (1)

In a particular climatic condition, decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and water-soluble substances like sugars.

[New NCERT Class 12th Page No. 208]

125. (1)

According to the International Union for Conservation of Nature and Natural Resources (IUCN) (2004), the total number of plant and animal species described so far is slightly more than 1.5 million.

[New NCERT Class 12th Page No. 217]

126. (4)

No energy that is trapped into an organism remains in it for ever.

[New NCERT Class 12th Page No. 209]

127. (4)

Ecology is basically concerned with four levels of biological organisation – organisms, populations, communities and biomes.

[New NCERT Class 12th Page No. 191]

128. (4)

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

[New NCERT Class 12th Page No. 199]

129. (1)

In the rocky intertidal communities of the American Pacific Coast the starfish *Pisaster* is an important predator. In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year, because of interspecific competition.

[New NCERT Class 12th Page No. 198]

130. (2)

Lichens represent an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria.

[New NCERT Class 12th Page No. 201]

131. (4)

- Bee and orchid – Mutualism
- Liver fluke – Endoparasite
- Barnacle and whale – Commensalism

[New NCERT Class 12th Page No. 200, 202]

132. (1)

A plot of N in relation to time (t) results in a sigmoid curve. This type of population growth is called *Verhulst-Pearl* Logistic Growth and is described by the following equation:

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

[New NCERT Class 12th Page No. 195]

133. (3)

Predators in nature are 'prudent'.

If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food.

[New NCERT Class 12th Page No. 198]

134. (3)

The size of a population for any species is not a static parameter.

It keeps changing with time, depending on various factors including food availability, predation pressure and adverse weather.

[New NCERT Class 12th Page No. 192]

135. (3)

The expression $(1-N/K)$ represents environmental resistance.

[New NCERT Class 12th Page No. 195]

136. (3)

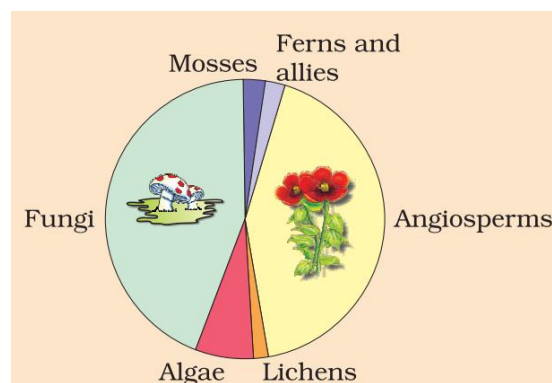
Many ecologists regard the entire biosphere as a global ecosystem, as a composite of all local ecosystems on Earth.

Crop fields and an aquarium may also be considered as man-made ecosystems,

[New NCERT Class 12th Page No. 205]

137. (2)

A-Fungi and B-Angiosperms.



[New NCERT Class 12th Page No. 218]

138. (4)

- It keeps prey population under control.
- It helps in maintaining species diversity.
- Biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate prey population.
- Predation is nature's way of transferring to higher trophic levels the energy fixed by plants.

[New NCERT Class 12th Page No. 197, 198]

139. (3)

Frog	Camouflage
Prickly pear cactus	Moth
Opium	Chemical defence
Thorns	Morphological defence

[New NCERT Class 12th Page No. 197, 198]

140. (2)

Secondary productivity is defined as the rate of formation of new organic matter by consumers.

[New NCERT Class 12th Page No. 207]

141. (1)

The rate of biomass production is called productivity. It is expressed in terms of $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$ to compare the productivity of different ecosystems.

[New NCERT Class 12th Page No. 207]

142. (3)

If N is the population density at time t , then its density at time $t + 1$ is:

$$N_{t+1} = N_t + [(B + I) - (D + E)].$$

[New NCERT Class 12th Page No. 193]

143. (4)

The cattle egret and grazing cattle show commensalism.

[New NCERT Class 12th Page No. 201]

144. (4)

Resource partitioning in which two species competing for the same resources can avoid competition by choosing. In this, the resources are

divided by the species within an ecological niche. It is an adaptation of interspecific competition.

[New NCERT Class 12th Page No. 199]

145. (2)

The given figure is of sigmoidal growth. It is a logistic/S-shaped growth curve. It is obtained due to the finite and limited source of resources in a population.

[New NCERT Class 12th Page No. 194]

146. (4)

India now has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries.

[New NCERT Class 12th Page No. 225]

147. (2)

There are more than 20,000 species of ants, 3,00,000 species of beetles, 28,000 species of fishes and nearly 20,000 species of orchids.

[New NCERT Class 12th Page No. 216]

148. (2)

Plants capture only 2-10 per cent of the PAR and this small amount of energy sustains the entire living world.

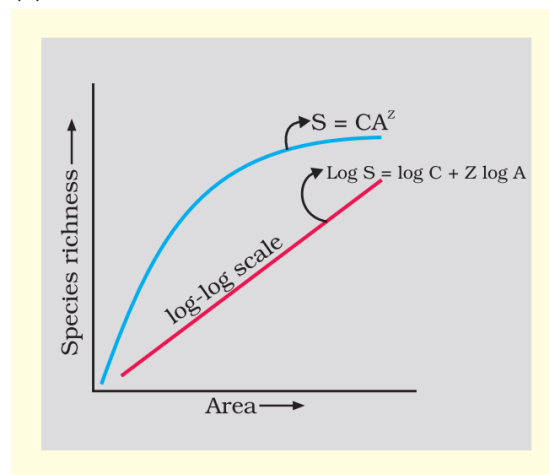
[New NCERT Class 12th Page No. 209]

149. (3)

Dodo	-	Mauritius
Quagga	-	Africa
Stellar's Sea Cow	-	Russia
Thylacine	-	Australia

[New NCERT Class 12th Page No. 221]

150. (2)



The following graph shows species area relationship and on log scale the relationship becomes linear.

[New NCERT Class 12th Page No. 220]

SECTION-(IV) ZOOLOGY

151. (2)

- Agriculture came around 10,000 years ago.
- *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like.
- *Homo erectus* had a large brain around 900cc.

[New NCERT Class 12th Page No. 124]

152. (4)

Taq DNA polymerase	Stable above 90°C
Exonuclease	Cleaves the ends of linear DNA
Protease	Degradation of protein
Chitinase	Breakdown of fungal cell wall

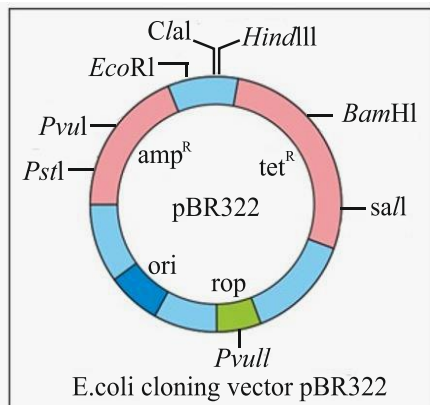
[New NCERT Class 12th Page No. 171]

153. (3)

Transgenic models exist for several human diseases, including Alzheimer's, to aid in the investigation of new treatments.

[New NCERT Class 12th Page No.183]

154. (4)



In pBR322,

- I- ori-represents site or origin of replication
- II- rop-codes for proteins that take part in the replication of plasmid.
- III- *EcoRI*- recognition sites of restriction endonuclease
- IV- *tet^R* – antibiotic resistance genes.

[New NCERT Class 12th Page No. 169]

155. (3)

- 'Cry' proteins are named so because they are crystal proteins.
- Transgenic mice are used to test vaccine safety.

[New NCERT Class 12th Page No. 208 & 209]

156. (2)

Vertebrate hearts and brains are homologous organs and show divergent evolution.

[New NCERT Class 12th Page No. 114]

157. (4)

- Restriction enzymes belong to a large class of enzymes called nuclease.
- *Hind II* cuts DNA at a specific sequence of six base pairs.
- Restriction enzymes cut the sugar-phosphate backbone of DNA.
- Restriction enzymes are named after the bacterial species they originate from and are defined by their ability to recognize specific, often palindromic, nucleotide sequences.

[New NCERT Class 12th Page No. 165]

158. (2)

- Traditional knowledge enhances biotechnological advancements and reduces research and development costs in agriculture and medicine.
- It primarily benefits developing countries and promotes biodiversity, making it valuable for scientific research.

[New NCERT Class 12th Page No. 184 -186]

159. (3)

Bobcat is a placental mammal and Koala is an Australian marsupial.

[New NCERT Class 12th Page No. 118]

160. (3)

The r-DNA molecule cannot be created, unless one cuts the DNA (source) and vector with same restriction enzyme, because when cut by same enzyme both will have same sticky ends, that can be joined by DNA ligase.

[New NCERT Class 12th Page No. 166]

161. (2)

Transgenic animals are designed to help researchers understand gene regulation and its impact on normal physiological functions and development.

[New NCERT Class 12th Page No. 183]

162. (2)

Insertion of gene of interest in selectable marker leads to insertional inactivation of selectable marker.

[New NCERT Class 12th Page No. 202]

163. (3)

- Special creation theory claims species exist in their current forms since creation.
- Natural selection is the process by which individuals better adapted to their environment survive and reproduce more successfully, leading to evolutionary changes.
- Panspermia theory proposes life spread through space; does not address earth's species.
- Abiogenesis theory explains life's origin from non-living matter; not about current forms.

[New NCERT Class 12th Page No. 112]

164. (1)

Bt toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides; in effect created a bio-pesticide.

[New NCERT Class 12th Page No. 179]

165. (3)

- RNAi is a biological process in which RNA molecules inhibit gene expression.
- It is used in plants for pest resistance by silencing specific genes, as in the case of nematode resistance in tobacco plants.

[New NCERT Class 12th Page No.180, 181]

166. (1)

H.M.S Beagle was the sail ship used by Charles Darwin during his famous voyage around the world.

[New NCERT Class 12th Page No.112]

167. (3)

- The recombinant DNA vector carries a gene for ampicillin resistance.
- When the *E. coli* cells are transformed with this vector and then spread on agar plates containing ampicillin, only those cells that have taken up the recombinant DNA and are thus resistant to ampicillin will survive and grow.
- Non-transformed cells, which do not have the ampicillin resistance gene, will be killed by the ampicillin and will not grow.

[New NCERT Class 12th Page No. 165]

168. (2)

Ramapithecus is an earlier hominid, followed by *Australopithecines*, then *Homo habilis*, and finally *Homo erectus*, which is more recent in the evolutionary timeline.

[New NCERT Class 12th Page No. 124]

169. (2)

Genetic Engineering Approval Committee is responsible for overseeing the safety of genetically modified organisms and ensuring that genetic engineering practices are safe for human health and the environment.

[New NCERT Class 12th Page No. 184]

170. (2)

- Bacterial cells are made 'competent' to take up DNA by treating them with divalent cations such as calcium chloride.
- This process increases the efficiency with which DNA enters the cells.
- Recombinant DNA can then be forced into such cells by incubating the cells with recombinant DNA on ice, followed by placing them briefly at 42°C (heat shock), and then putting them back on ice. This enables the bacteria to take up the recombinant DNA.

[New NCERT Class 12th Page No. 200]

171. (2)

In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B, chains of human insulin, and introduced them in plasmids of *E. coli* to produce insulin chains. Chains A and B were produced separately, extracted, and combined by creating disulphide bonds to form human insulin.

[New NCERT Class 12th Page No. 212]

172. (1)

Genetically Modified Organisms (GMOs) are plants, bacteria, fungi, and animals with altered genes. GM plants have benefited by:

- Reducing reliance on chemical pesticides. Minimising post-harvest losses.
- Enhancing mineral usage efficiency, preserving soil fertility. Improving nutritional value, e.g., Vitamin A-enriched golden rice.
- It has made crops more tolerant to abiotic stresses (cold, drought, salt, heat).

[New NCERT Class 12th Page No. 179]

173. (3)

The main objective of inserting alien DNA into a cloning vector and transferring it into a host cell is to express and produce a desirable protein. This involves creating recombinant DNA that directs the host cell to make a specific protein.

[New NCERT Class 12th Page No. 173]

174. (2)

- The amendment aims to prevent unauthorized use of India's bio-resources and ensure fair compensation for local communities.
- Targets exploitation, not just agricultural output.
- Regulates domestic bio-resources, not foreign patents.
- Seeks to protect traditional knowledge, not ban it.

[New NCERT Class 12th Page No. 184-185]

175. (4)

Methylation	Addition of methyl group
1963	Year when the enzymes restricting bacteriophage growth were discovered.
Bacteriophage	Virus that infects bacteria.
Elution	Extraction of DNA from the gel plate

[New NCERT Class 12th Page No. 201]

176. (2)

The geological history of earth correlates closely with biological history.

[New NCERT Class 12th Page No. 113]

177. (3)

The Big Bang theory describes a singular explosive event leading to the formation of the universe. Continuous expansion and the formation of stars are later consequences, not the main event.

[New NCERT Class 12th Page No. 111]

178. (3)

- DNA is a negatively charged molecule, so they can be separated by forcing them to move towards the anode under an electric field.
- DNA fragments separate according to size through the pores of agarose gel.
- The separated DNA fragments can be visualized only after staining the DNA with a compound known as ethidium bromide then followed by exposure to UV radiation.

[New NCERT Class 12th Page No. 198]

179. (3)

- Antisense technology is used to control the gene expression by using RNA molecule which is complementary to the DNA sequence.
- This RNA molecule binds to the DNA and stops the translation process and ultimately stops protein synthesis.
- So, this technology is used to stop the synthesis of a particular protein by targeting a specific gene.

[New NCERT Class 12th Page No. 209]

180. (3)

Dark winged moth survived after industrialisation. The tree trunks became dark due to industrial smoke and soot due to which dark winged moth easily camouflage.

[New NCERT Class 12th Page No. 116]

181. (3)

- Pasteur conducted experiments using swan-neck flasks, which allowed air in but prevented microbial contamination.
- His findings showed that no microbial life developed in the flasks unless they were exposed to pre-existing microorganisms, thus disproving spontaneous generation.

[New NCERT Class 12th Page No.111]

182. (3)

The main challenge for production of insulin using rDNA techniques was getting insulin assembled to mature form.

[New NCERT Class 12th Page No. 187]

183. (2)

Genetic modifications can lead to unintended ecological consequences, such as the disruption of local ecosystems.

[New NCERT Class 12th Page No.186]

184. (2)

- Asexual reproduction preserves the genetic information, while sexual reproduction permits variation.
- Each restriction endonuclease functions by 'inspecting' the length of a DNA sequence.

[New NCERT Class 12th Page No. 194]

185. (3)

- Both early diagnosis and understanding the disease's mechanism are essential for effective treatment.
- Either diagnosis or understanding alone is insufficient for proper treatment.
- Recombinant DNA technology, Polymerase Chain Reaction (PCR), and Enzyme-Linked Immuno-sorbent (ELISA) are some of the techniques that serve the purpose of early diagnosis.

[New NCERT Class 12th Page No. 183]

186. (3)

Orange coloured bands of DNA can be visualized if an ethidium bromide-stained gel exposed to UV light.

[New NCERT Class 12th Page No. 168]

187. (2)

The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase.

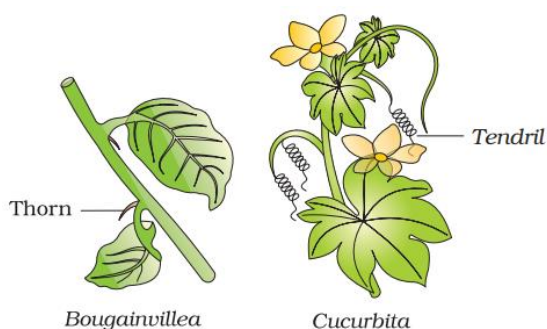
[New NCERT Class 12th Page No. 182]

188. (1)

The *Bt* toxin protein exists as inactive protoxins but once an insect ingests the inactive toxin, it is converted into an active form of toxin due to the alkaline pH of the gut which solubilises the crystals. The activated toxin binds to the surface of midgut epithelial cells and creates pores that cause cell swelling and lysis and eventually cause death of the insect.

[New NCERT Class 12th Page No. 208 & 209]

189. (2)



The thorn and tendrils of *Bougainvillea* and *Cucurbita* represent homology and are result of divergent evolution.

[New NCERT Class 12th Page No. 115]

190. (2)

- Bacteriophages have high copy numbers in the bacterial cells.
- The first step in genetic modification is identifying DNA with useful genes.

[New NCERT Class 12th Page No. 198]

191. (2)

The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill, that takes such issues into consideration, including patent terms emergency provisions and research and development initiative.

[New NCERT Class 12th Page No. 185]

192. (4)

- Darwin's finches are small black birds.
- The brain capacities of *Homo habilis* were between 650-800cc.
- *Ichthyosaurs* was fish like reptile.
- The original finches were seed-eating.

[New NCERT Class 12th Page No. 118]

193. (4)

- PCR is used for amplifying DNA, detecting pathogens, and identifying gene mutations, but it is not used directly for isolating genes for recombinant purposes.
- PCR is used for early pathogen detection, identifying mutations, and detecting HIV.

[New NCERT Class 12th Page No. 183]

194. (2)

<i>PvuI</i>	amp ^R
PCR	Amplification of genetic material
Ligase	Join the sticky ends of DNA fragments
Ti plasmid	<i>Agrobacterium</i>

[New NCERT Class 12th Page No. 166]

195. (2)

The embryos of all vertebrates including human develop a row of vestigial gill slit just behind the head. This shows embryological evidence.

[New NCERT Class 12th Page No. 113]

196. (1)

Presence of insert results into insertional inactivation of the β -galactosidase and the colonies do not produce any colour, these are identified as recombinant colonies.

[New NCERT Class 12th Page No. 200]

197. (4)

The gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages, which could be a permanent cure for ADA deficiency.

[New NCERT Class 12th Page No. 211]

198. (1)

- The tumor-inducing plasmid (Ti Plasmid) acts as a cloning vector in recombinant DNA technology.

- The Ti plasmid used in gene delivery mechanisms is disarmed, meaning it no longer remains pathogenic.

[New NCERT Class 12th Page No. 170]

199. (2)

- Darwin's theory emphasises natural selection, where characteristics that aid survival influence reproductive success.
- Darwin's theory rejects static species and affirms common ancestry.
- Extinction is a part of natural selection.

[New NCERT Class 12th Page No.113]

200. (3)

To recover many copies of target DNA, it should be cloned in a vector with a high-copy-number origin.

[New NCERT Class 12th Page No. 169]

